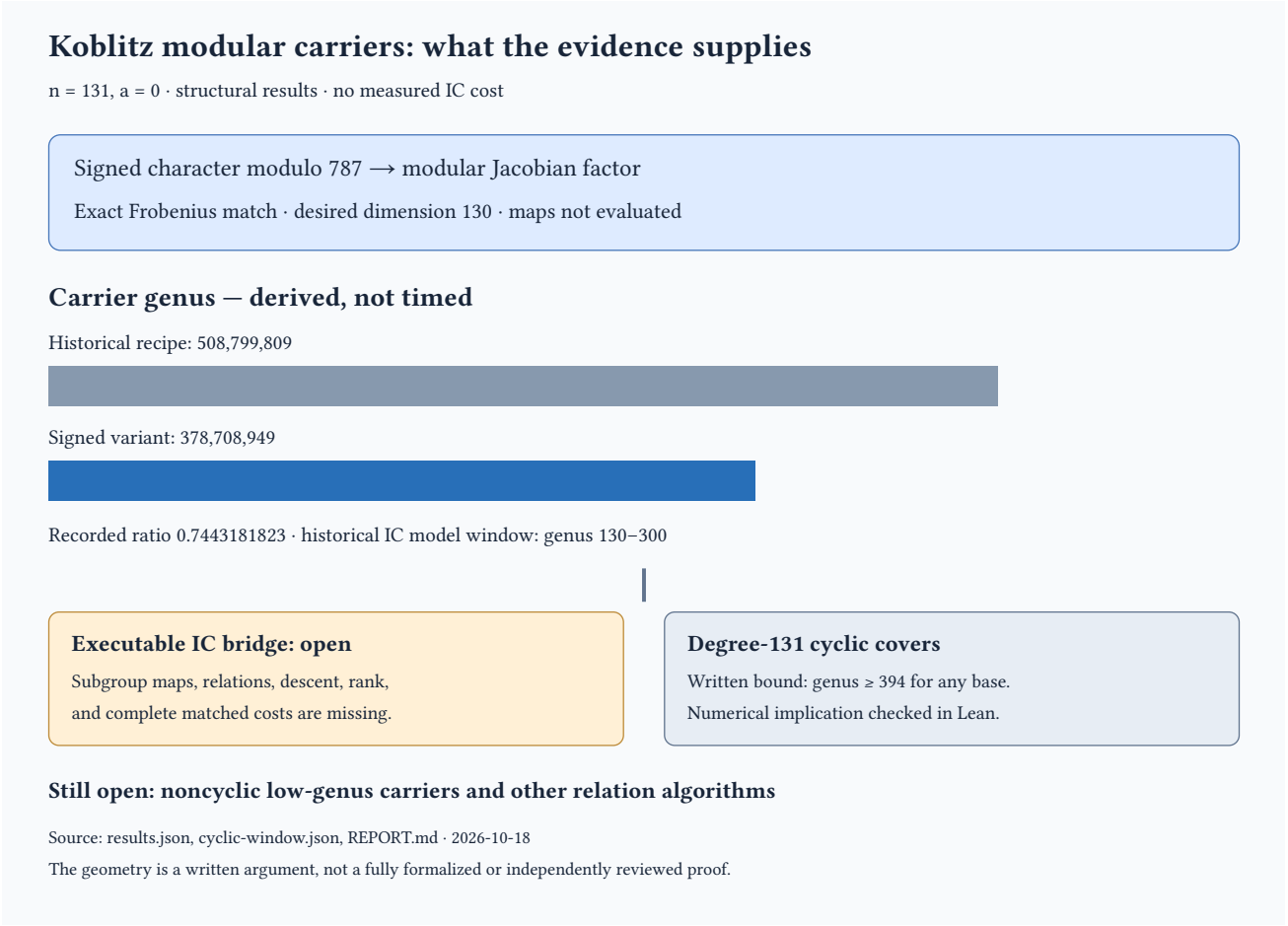
Koblitz curves, modular carriers, and index calculus

2026-10-18. Structural investigation of the supplied Klein-quartic and modular-twist lead. All new computation is native Rust; no DLP was run.

The lead is mathematically real, and the repository already explored it in its curve-construction study. This investigation adds a smaller signed-character modular carrier and a written extension of the cyclic-cover obstruction. Neither supplies an executable improvement to the current index-calculus pipeline.



Scope and exact target

The source is E_0: y^2+xy=x^3+1 over F_(2^131), with prime subgroup r=6805647338418769269323201294934099851 29, cofactor 4, and registry slug icv1-f2m131-tm22283658519494248867-115e0dc5. Its field modulus is 0x80000000000000000000000000000002007. The registry leaves its EC1 generator binding unresolved. This work uses the family, Frobenius, and subgroup order; it imports no challenge point and runs no unknown-scalar recovery.

Let A_n(E) be the connected trace-zero factor of Res_(F_(2^n)/F_2)(E). For n=131 its dimension is 130. At F_2, #A_131(E_0)=r exactly. Polynomial conventions below use P(X)=det(X-Frob); L(T)=T^(2g)P(1/T).

1. What is established about the original lead

For C: x^3 y+y^3 z+z^3 x=0 over F_2, the replay gives point counts 3,5,24,17,33,38,129,257 over extensions of degrees 1 through 8. The first three counts and reciprocity determine

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L_C(T)=1+5T^3+8T^6.
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This identifies J(C), up to F_2-isogeny, with Res_(F_8/F_2)(E_1). It agrees with the classical characteristic-2 Klein-quartic picture in Elkies. An isogeny class is not an evaluated subgroup transfer. In particular, the obvious F_2-rational cyclic quotient to E_1 detects the elliptic factor; it kills the trace-zero factor. Obtaining the trace-zero transfer needs other correspondences, rather than that quotient alone.

For an odd prime n !=7, take the conductor-49 CM newform f with a_2(f)=1 and an odd-conductor character chi whose value chi(2) is primitive of order n. The conjugate Frobenius roots of f tensor chi are alpha*zeta^j and beta*zeta^j, 1<=j<n, where alpha+beta=1, alpha*beta=2. These are precisely the roots of P_(A_n(E_1)). This uses the standard twisting and modular-abelian-variety constructions, not a new modularity theorem. See Atkin--Li and Stein's modular abelian variety construction. The exception n=7 must be handled separately: the CM field already lies in Q(zeta_7), and the trace-zero factor splits, as the repository observed.

For composite n, one primitive-order twist generally supplies only the primitive cyclotomic piece. At n=9, the full trace-zero dimension is 8, the primitive-order-9 piece has dimension 6, and the missing order-3 piece has dimension 2. Both the character's

value at 2 and the lower-order pieces must be accounted for. The broad existence construction can use multiple twists and their common modular level; one order-n twist is not enough in general.

2. New construction: absorb the sign into the character

The historical E_0 construction first twists f by the quadratic character of conductor 3, then by an order-131 character modulo 263. Its level is 441*263^2=30,503,529 and its carrier genus is 508,799,809.

Instead choose chi modulo 787 with

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chi(2)=exp(2*pi*i/262).
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The native check proves that 787 is prime and 2 is a primitive root modulo 787 (by factoring 786 and testing its prime divisors). Thus chi is primitive of order 262. Every embedding of Q(zeta_262)=Q(zeta_131) takes chi(2) to a primitive 262nd root, equivalently minus a primitive 131st root. The Frobenius roots are therefore minus the E_1 roots times all primitive 131st roots: exactly A_131(E_0).

The coefficient field is exactly Q(zeta_131): it is contained there because f has rational coefficients, and a_2(f tensor chi)=chi(2) generates it. Consequently the attached modular abelian variety has dimension 130. Since 787 is coprime to 49, the twist has level 49*787^2=30,349,081. Its nebentypus chi^2 has order 131, so it occurs on the character-kernel curve X_H(N), H=ker(chi^2), of degree 131 over X_0(N).

The character chi is odd. This is permitted: the weight-2 form's nebentypus is chi^2, which is even. Requiring chi itself to be even would miss the construction.

Structural quantity	Historical E_0 recipe	Signed-character variant
Character modulus	263, plus separate sign at 3	787
Character order at 2	131, plus sign	262
Modular level	30,503,529	30,349,081
Index over X_0	131	131
Genus of X_0	3,883,969	2,890,909
Genus of carrier X_H	508,799,809	378,708,949
Dimension of desired factor	130	130

The carrier-genus ratio is 0.7443181823207013, a reduction of about 25.57%. This is a genus reduction, not a speed measurement or an IC improvement. The new genus is still over 1.26 million times 300. The scan covers prime moduli <=10,000, excluding the overlapping conductor 7; it establishes the best result within that family and bound, not the smallest possible carrier among all curves or modular quotients.

For n=5 the same device uses an order-10 character modulo 11: it realizes the E_0 factor at genus 2,841 instead of the old genus 36,001. This is a useful control, while the repository's direct-search questions at much smaller genus remain separate.

Genus calculation and verification

For N=49*787^2, mu_0=56*787*788=34,728,736, c_0=8*788=6,304, e_2=0, e_3=4. The degree-131 character cover is unramified over cusps: at every cusp denominator c|N the lower-right entry of its parabolic stabilizer is 1 modulo 787, because 787 divides c*N/gcd(c^2,N). It is also unramified over elliptic points: their stabilizer orders are 2 or 3, coprime to 131. Hence

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g_H=1+131*(g_0-1)=378,708,949.
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This proves the split-cusp assertion for this construction rather than inferring it only from small tests. The genus formula is standard; see Snowden's derivation. Complete coset controls at six small levels check generalized kernels, including kernels larger than {+1,-1} and an order-3 elliptic-ramification case. No full-size coset enumeration was attempted.

The Frobenius calculation is checked coefficient by coefficient over Z, using two constructions: monic division of X^(2n)-s_n X^n+2^n, and the quadratic norm of sum(alpha^j X^(n-1-j),j=0..n-1). Twenty checks cover both trace signs at n=3,5,7,11,13,41,53,61,83,131. This replaces the historical large-target floating-point evaluation check with an exact one. The full-size identity certificate is IDC1h650837b4b977a1d0: 64 issuer points and 128 additional checker points, plus an independently implemented quadratic norm versus monic-division coefficient comparison in the native driver. The two arithmetic constructions are distinct; the replay remains same-agent/same-host rather than an independent validation.

3. New written bound: no degree-131 cyclic carrier below genus 394

Statement. Let C and B be smooth projective geometrically connected curves over F_2, and C->B a finite separable map, geometrically Galois with cyclic group of order 131. If J(C) has A_131(E_0) as an F_2-isogeny factor, then g(C)>=394. The same argument applies to E_1. This is a written mathematical argument with an arithmetic replay and conditional Lean proof; the geometric bridge is not formalized or independently reviewed.

Proof. Put g=g(C), b=g(B), and let R count geometric branch points. The cover is tame, and every ramified point is totally ramified. Therefore

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g=131*(b-1)+1+65*R.                                    (1)
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Let d|130 be the order of Frobenius acting on the geometric deck group. On the Prym, Frobenius permutes the 130 nontrivial character spaces in cycles of length d, so its eigenvalue multiset is stable under mu_d. The roots of A are alpha*zeta^j and beta*zeta^j, 1<=j<=130. The alpha/beta ratio is not a root of unity (the elliptic curve is ordinary), and gcd(d,131)=1. Their mu_d orbit therefore has 260d distinct elements. For b<130, simplicity of A puts it in the Prym, giving

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g-b >=130*d.                                    (2)
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For completeness, simplicity follows because pi=alpha*zeta_131 generates Q(sqrt(-7),zeta_131), of degree 260: pi^131 is a nonrational element of Q(sqrt(-7)), so alpha and then zeta_131 can be recovered in Q(pi). The two constituent fields are disjoint. This uses the ordinary ratio condition to exclude rationality of alpha^131.

Suppose g<=393. Equation (1) gives b<=3. Equation (2), together with d|130, gives d=1 or 2. The deck group becomes constant over F_(2^d). Every abelian variety D of dimension b over F_2 satisfies #D(F_2)<=5^b. One proof writes its real Weil roots as lambda_i. The positive algebraic integer product prod(3+lambda_i) is >=1, so AM--GM gives mean(lambda_i)>=-2; a second AM--GM gives prod(3-lambda_i)<=5^b. This is also a special case of Aubry--Haloui--Lachaud, Corollary 2.2. The quadratic twist D^- obeys the same bound, and

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#D(F_4)=#D(F_2)*#D^-(F_2).
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For b<=3 each factor is <=125<131. Thus J(B)(F_(2^d)) has no 131-torsion. Unramified geometric cyclic degree-131 covers over this field are therefore excluded by geometric class field theory. See Conrad's geometric CFT notes. There must be a branch place. Since the extension has constant cyclic Galois group and is tame, its residue field contains mu_131. Since ord_131(2)=130, that place has degree at least 130/d over F_(2^d). Consequently R>=130/d>=65. Equation (1) now gives g>=4095, a contradiction. Thus g>=394. QED.

The arithmetic replay finds 30 tuples satisfying (1) and (2) below genus 394; all are rejected by this branch constraint. It enumerates integer constraints, not curves. The bound does not assert existence at genus 394. The earlier >=1300 bound for bases of genus <=1 remains stronger in its smaller family. The new statement extends the exclusion to every base through genus 393, closing this particular cyclic mechanism throughout the historical genus-130--300 IC window.

4. What remains for index calculus

Neither an abstract modular factor nor a small dimension supplies a smoothness oracle. The factor dimension 130 must not be substituted for the carrier genus 378,708,949 in a Jacobian smoothness estimate. Conversely, large ambient genus alone is not a theorem against every projected or modular-specific relation algorithm. Such an algorithm remains unconstructed.

The historical genus window and log2 cost 37.17 at genus 130 are model outputs, not measured recovery costs. A small carrier would still need validated transfer, rank and relation-distribution tests, individual descent, and complete setup/online/recovery pricing before an IC claim.

Requirement	Status	Evidence or missing bridge
Audit the Klein and modular claims	Verified at stated polynomial/ construction scope	Eight Klein counts; exact prime-degree polynomials; composite control
Seek a smaller modular carrier	Verified structural result in scanned family	Signed character modulo 787; genus proof and controls
Test the direct cyclic generalization	Written bound plus checked numerical implication	g>=394 under the exact hypotheses above
Explicit curve equations and evaluated subgroup maps at n=131	Not implemented	No Hecke projector, kernel certificate, or target lift evaluated
Faster IC on the projected factor	Not implemented or measured	No modular relation/decomposition algorithm, descent, rank or cost data
Noncyclic low-genus carrier	Open	No existence or nonexistence proof in genus 130--300
Full formal proof of geometry and modularity	Partial	Only arithmetic implication is in Lean; see proof map

The next discriminating work is a low-genus carrier outside the excluded degree-131 cyclic family: noncyclic/non-Galois constructions, gluing and polarization on A times a small cofactor, or a modular quotient with verified action preserving A. A quotient must preserve the desired factor; its existence cannot be inferred from an abelian-variety quotient alone. The n=5 rung is a useful first test. The n=7 carriers are positive controls, but A_7 splits into two three-dimensional factors, so that rung cannot stand in for the simple n=131 case. See the repository's Torelli toy study.

Keep the missing executable-transfer and relation-algorithm obligations in the problem record.

5. Reproduction, proof map, and reporting limits

See PROTOCOL.md, results.json, cyclic-window.json, native source, and VALIDATION.md. First-run outputs are preserved.

```
cargo test --release --offline --locked --manifest-path research/koblitz_modular_ic_20261018/native/Cargo.toml
mkdir /tmp/koblitz-carrier-replay
cargo run --release --offline --locked --manifest-path research/koblitz_modular_ic_20261018/native/Cargo.toml -- /tmp/koblitz-
carrier-replay
cmp research/koblitz_modular_ic_20261018/results.json /tmp/koblitz-carrier-replay/results.json
cargo run --release --offline --locked --manifest-path research/koblitz_modular_ic_20261018/native/Cargo.toml -- --cyclic-only /
tmp/koblitz-carrier-replay/cyclic-window.json
cd research/koblitz_modular_ic_20261018
lean CyclicWindow.lean
```

Mathematical step	Lean declaration / verification
Numerical cyclic genus implication	cyclic_window_numeric; kernel checked with Lean 4.34.1; only propext, Classical.choice, Quot.sound
5^3=125<131	dimension_three_order_bound; kernel checked without axioms
Curve/Prym/Frobenius and CFT bridge	Written argument above; not formalized
Signed modular carrier and exact genus	Written argument plus native exact checks; not formalized
Trace-zero polynomial identity	Native coefficient equality plus identity.certificate/v1 records; evaluation certificates alone prove only their finite-field scope

No custom axioms, sorry, native-decide bypasses, or assumed genus conclusions occur in the successful Lean file. An initial failed elaboration log is preserved separately and is not a proof. Full formalization needs the finite-field curve, abelian-variety, Prym-character and CFT dependencies; the conditional arithmetic theorem does not replace them.

Graph review: the canonical scoreboard and current-state summary update the carrier genus and scoped cyclic bound. Their performance values, the progress timeline and IC leaderboard remain unchanged: this study has no measured cost, S, speedup, or admitted performance row. The new diagram and PDF record the structural findings without changing those values. All checks here are same-agent/same-host, not independent-host validation.